

and business opportunity in the current energy situation.” The CGAI was set up in 2022 to promote the national mission.

Balyan sees huge potential if the 100 MT coal gasification target is achieved: “It will take care of 15-20% of India’s need for methanol, ammonium nitrate and SNG; liquid fuel and other chemicals can follow.” Although he is not sure if this would indeed happen, he says the stage is set for a take-off.

WHY INDIA HAS TRAILED

India’s tryst with coal gasification began with fertiliser plants at Sindri, Talcher and Ramagundam in 1970s-’80s using imported gasifiers. But they were shut between 1999 and 2002 due to financial non-viability. The Sindri and Ramagundam plants were rebuilt in 2021-22 using imported LNG. The Talcher plant is being revived using US-based Air Products’ coal gasifier.

With easy availability of cheap imported LNG and LPG, India’s preference changed but coal gasification wasn’t abandoned.

70 | Debajit Palit, who heads the Chintan Research Foundation (CRF), points out that India continued to use the gasification route to power and heat generation, but not for chemicals and liquid fuels. “We never thought of a crisis in supply of LNG and LPG,” he adds.

That scenario changed recently, after two big private players went for coal gasification.

In 2018, Jindal Steel’s Angul plant in Odisha started producing electricity using a South African gasifier. It is now expanding the facility with the Centre’s VGF.

That year, Reliance Industries’ Jamnagar refinery started producing electricity and hydrogen using Houston-based Phillips 66’s gasifier.

But India couldn’t match the miracle of China and South Africa.

China and South Africa also adopted gasification with imported technology in the 1950s. But they innovated and indigenised the technology. China now produces 90% of its own and 30% of global ammonia and 80% of its own and 40% of global urea, some of which is exported to India. It also produces 70% of its methanol and other petrochemicals. South Africa produces 30% of synthetic liquid fuels like petrol, diesel, kerosene, LPG, and naphtha—apart from gas and petrochemicals.

Imagine had India pursued coal gasification as relentlessly. The West Asia war wouldn’t be hurting growth, triggering reverse migration or bringing workers and housekeepers to streets.

CENTRE’S INCENTIVES

- 1 ₹8,500 CRORE VIABILITY GAP FUNDING (VGF) FOR PUBLIC AND PRIVATE SECTOR FIRMS
- 2 APPROVED INVESTMENT BY COAL INDIA (CIL) IN JOINT VENTURES WITH OTHER PSUs
- 3 NEW SUB-SECTOR, ‘PRODUCTION OF SYNGAS LEADING TO COAL GASIFICATION,’ LAUNCHED; TO ENSURE CHEAPER COAL
- 4 50% REBATE IN REVENUE SHARE FOR COAL USED, PROVIDED AT LEAST 10% OF TOTAL COAL PRODUCTION IS USED FOR GASIFICATION

THE VGF MATH

- 1 **TOTAL BUDGETED VGF:** ₹8,500 CRORE
- 2 **CATEGORY-I (CPSU PROJECTS):** ₹4,050 CRORE; LUMP SUM ₹1,350 CRORE, OR 15% OF CAPEX, WHICHEVER IS LOWER
- 3 **CATEGORY-II (PUBLIC OR PRIVATE PROJECTS):** ₹3,850 CRORE; LUMP SUM ₹1,000 CRORE, OR 15% OF THE CAPEX, WHICHEVER IS LOWER
- 4 **CATEGORY-III (DEMO OR SMALL PROJECTS USING INDIGENOUS GASIFIERS):** ₹600 CRORE; LUMP SUM ₹100 CRORE, OR 15% OF CAPEX, WHICHEVER IS LOWER

